NumPy Broadcasting & JAX

Lecture 07

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Basic file IO

Reading and writing ndarrays

We will not spend much time on this as most data you will encounter is more likely to be a tabular format (e.g. data frame) and tools like Pandas are more appropriate.

For basic saving and loading of NumPy arrays there are the `save()` and `load()` functions which use a custom binary format.

```
x = np.arange(1e5)
np.save("data/x.npy", x)
```

```
new_x = np.load("data/x.npy")
np.all(x == new_x)
```

```
np.True_
```

Additional functions for saving (`savez()`, `savez_compressed()`, `savetxt()`) exist for saving multiple arrays or saving a text representation of an array.

Reading delimited data

While not particularly recommended, if you need to read delimited (csv, tsv, etc.) data into a NumPy array you can use `genfromtxt()`,

```
with open("data/mtcars.csv") as file:
    mtcars = np.genfromtxt(file, delimiter=",", skip_header=True)

mtcars
```

```
array([[  6.   , 160.   , 110.   ,   3.9  ,   2.62 ,
         16.46 ,   0.   ,   1.   ,   4.   ,   4.   ],
       [  6.   , 160.   , 110.   ,   3.9  ,   2.875,
         17.02 ,   0.   ,   1.   ,   4.   ,   4.   ],
       [  4.   , 108.   ,  93.   ,   3.85 ,   2.32 ,
         18.61 ,   1.   ,   1.   ,   4.   ,   1.   ],
       [  6.   , 258.   , 110.   ,   3.08 ,   3.215,
         19.44 ,   1.   ,   0.   ,   3.   ,   1.   ],
       [  8.   , 360.   , 175.   ,   3.15 ,   3.44 ,
         17.02 ,   0.   ,   0.   ,   3.   ,   2.   ],
       [  6.   , 225.   , 105.   ,   2.76 ,   3.46 ,
         20.22 ,   1.   ,   0.   ,   3.   ,   1.   ],
       [  8.   , 360.   , 245.   ,   3.21 ,   3.57 ,
         15.84 ,   0.   ,   0.   ,   3.   ,   4.   ],
       [  4.   , 146.7  ,  62.   ,   3.69 ,   3.19 ,
         20.   ,   1.   ,   0.   ,   4.   ,   2.   ],
```

```
[  4.   , 140.8  ,  95.   ,   3.92 ,   3.15 ,
  22.9  ,   1.   ,   0.   ,   4.   ,   2.   ],
[  6.   , 167.6  , 123.   ,   3.92 ,   3.44 ,
```

Broadcasting

Broadcasting

This is an approach for deciding how to generalize operations between arrays with differing shapes.

```
x = np.array([1, 2, 3])
```

```
x * 2
```

```
array([2, 4, 6])
```

```
x * np.array([2,2,2])
```

```
array([2, 4, 6])
```

```
x * np.array([2])
```

```
array([2, 4, 6])
```

Efficiency

Using broadcasts can be more efficient as it does not copy the broadcast data,

```
x = np.arange(1e5)
y = np.array([2]).repeat(1e5)
```

```
%timeit x * 2
```

```
13.1 µs ± 297 ns per loop (mean ± std. dev. of 7 runs, 100,000 loops each)
```

```
%timeit x * np.array([2])
```

```
18.3 µs ± 96.4 ns per loop (mean ± std. dev. of 7 runs, 10,000 loops each)
```

```
%timeit x * y
```

```
61.4 µs ± 465 ns per loop (mean ± std. dev. of 7 runs, 10,000 loops each)
```

```
%timeit x * np.array([2]).repeat(1e5)
```

```
99.2 µs ± 1.12 µs per loop (mean ± std. dev. of 7 runs, 10,000 loops each)
```

Rules for Broadcasting

> When operating on two arrays, NumPy compares their shapes element-wise. It starts with the trailing (i.e. rightmost) dimensions and works its way left. Two dimensions are compatible when
>
> 1. they are equal, or
> 2. one of them is 1
>
> If these conditions are not met, a `ValueError: operands could not be broadcast together` exception is thrown, indicating that the arrays have incompatible shapes. The size of the resulting array is the size that is not 1 along each axis of the inputs.

Example

Why does the code on the left work but not the code on the right?

```
x = np.arange(12).reshape((4,3)); x
```

```
array([[ 0,  1,  2],
       [ 3,  4,  5],
       [ 6,  7,  8],
       [ 9, 10, 11]])
```

```
x + np.array([1,2,3])
```

```
array([[ 1,  3,  5],
       [ 4,  6,  8],
       [ 7,  9, 11],
       [10, 12, 14]])
```

```
x = np.arange(12).reshape((3,4)); x
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
```

```
x + np.array([1,2,3])
```

```
ValueError: operands could not be broadcas
```

```
x    (2d array): 4 x 3
y    (1d array):     3
----------------------
x+y  (2d array): 4 x 3
```

```
x    (2d array): 3 x 4
y    (1d array):     3
----------------------
x+y  (2d array): Error
```

A fix

```
x = np.arange(12).reshape((3,4)); x
```

```
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
```

```
x + np.array([1,2,3]).reshape(3,1)
```

```
array([[ 1,  2,  3,  4],
       [ 6,  7,  8,  9],
       [11, 12, 13, 14]])
```

```
x    (2d array): 3 x 4
y    (2d array): 3 x 1
---------------------
x+y  (2d array): 3 x 4
```

Examples (2)

```
x = np.array([0,10,20,30]).reshape((4,1))
y = np.array([1,2,3])
```

```
x
```

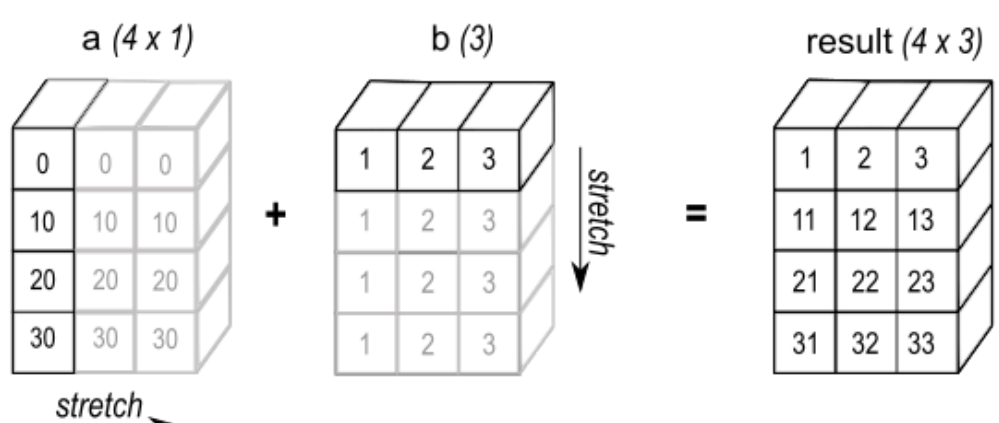
```
array([[ 0],
       [10],
       [20],
       [30]])
```

```
x+y
```

```
array([[ 1,  2,  3],
       [11, 12, 13],
       [21, 22, 23],
       [31, 32, 33]])
```

```
y
```

```
array([1, 2, 3])
```



Exercise 1

For each of the following combinations determine what the resulting dimension will be using broadcasting

- A [128 x 128 x 3] + B [3]
- A [8 x 1 x 6 x 1] + B [7 x 1 x 5]
- A [2 x 1] + B [8 x 4 x 3]
- A [3 x 1] + B [15 x 3 x 5]
- A [3] + B [4]

Demo 1 - Standardization

Below we generate a data set with 3 columns of random normal values. Each column has a different mean and standard deviation which we can check with `mean()` and `std()`.

```
rng = np.random.default_rng(1234)
d = rng.normal(
  loc=[-1,0,1],
  scale=[1,2,3],
  size=(1000,3)
)
```

```
d.shape
```

```
(1000, 3)
```

```
d.mean(axis=0)
```

```
array([-1.02944, -0.01396,  1.01242])
```

```
d.std(axis=0)
```

```
array([0.99675, 2.03223, 3.10625])
```

Lets use broadcasting to standardize all three columns to have mean 0 and standard deviation 1.

Broadcasting and assignment

In addition to arithmetic operators, broadcasting can be used with assignment via array indexing,

```
x = np.arange(12).reshape((3,4))
y = -np.arange(4)
z = -np.arange(3)
```

```
x[:] = y
x
```

```
array([[ 0, -1, -2, -3],
       [ 0, -1, -2, -3],
       [ 0, -1, -2, -3]])
```

```
x[...] = y
x
```

```
array([[ 0, -1, -2, -3],
       [ 0, -1, -2, -3],
       [ 0, -1, -2, -3]])
```

```
x[:] = z
```

```
ValueError: could not broadcast input arra
```

```
x[:] = z.reshape((3,1))
x
```

```
array([[ 0,  0,  0,  0],
       [-1, -1, -1, -1],
       [-2, -2, -2, -2]])
```

JAX

JAX

> JAX is a library for array-oriented numerical computation (à la NumPy), with automatic differentiation and JIT compilation to enable high-performance machine learning research.
>
> - JAX provides a unified NumPy-like interface to computations that run on CPU, GPU, or TPU, in local or distributed settings.
> - JAX features built-in Just-In-Time (JIT) compilation via Open XLA, an open-source machine learning compiler ecosystem.
> - JAX functions support efficient evaluation of gradients via its automatic differentiation transformations.
> - JAX functions can be automatically vectorized to efficiently map them over arrays representing batches of inputs.

```python
import jax
jax.__version__
```

```
'0.9.0'
```

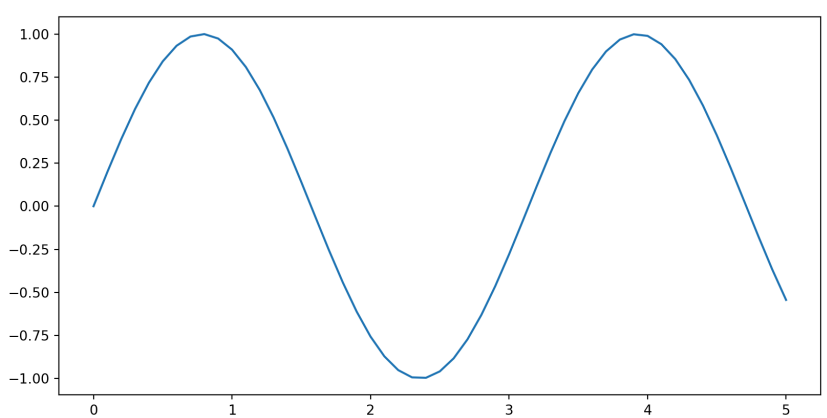
JAX & NumPy

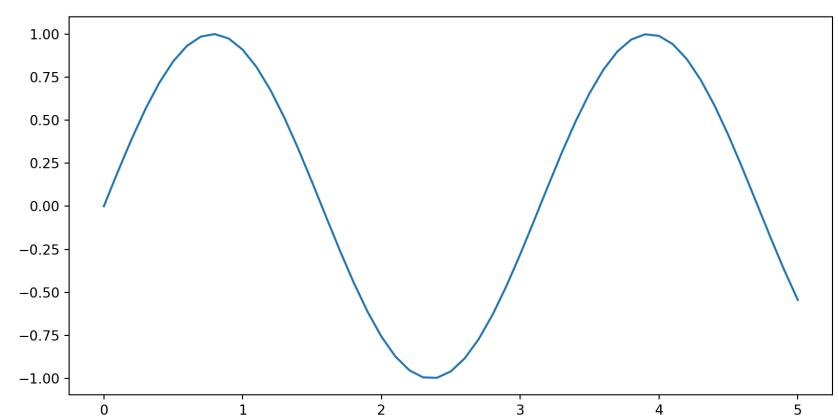
```python
import numpy as np

x_np = np.linspace(0, 5, 51)
y_np = 2 * np.sin(x_np) * np.cos(x_np)
plt.plot(x_np, y_np)
```

```python
import jax.numpy as jnp

x_jnp = jnp.linspace(0, 5, 51)
y_jnp = 2 * jnp.sin(x_jnp) * jnp.cos(x
plt.plot(x_jnp, y_jnp)
```






```python
type(x_np)
```

```
<class 'numpy.ndarray'>
```

```python
type(x_jnp)
```

```
<class 'jaxlib._jax.ArrayImpl'>
```

```python
x_np
```

```
array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6,
       1. , 1.1, 1.2, 1.3, 1.4, 1.5, 1.6,
       2. , 2.1, 2.2, 2.3, 2.4, 2.5, 2.6,
       3. , 3.1, 3.2, 3.3, 3.4, 3.5, 3.6,
       4. , 4.1, 4.2, 4.3, 4.4, 4.5, 4.6,
       5. ])
```

```python
x_jnp
```

```
Array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6,
       1. , 1.1, 1.2, 1.3, 1.4, 1.5, 1.6,
       2. , 2.1, 2.2, 2.3, 2.4, 2.5, 2.6,
       3. , 3.1, 3.2, 3.3, 3.4, 3.5, 3.6,
       4. , 4.1, 4.2, 4.3, 4.4, 4.5, 4.6,
       5. ], dtype=float32)
```

```python
x_np.dtype
```

```
dtype('float64')
```

```python
x_jnp.dtype
```

```
dtype('float32')
```

Compatibility

```
y_mix = 2 * np.sin(x_jnp) * jnp.cos(x_np); y_mix
```

```
Array([ 0.     ,  0.19867,  0.38942,  0.56464,  0.71736,
        0.84147,  0.93204,  0.98545,  0.99957,  0.97385,
        0.9093 ,  0.8085 ,  0.67546,  0.5155 ,  0.33499,
        0.14112, -0.05837, -0.25554, -0.44252, -0.61186,
       -0.7568 , -0.87158, -0.9516 , -0.99369, -0.99616,
       -0.95892, -0.88345, -0.77276, -0.63127, -0.4646 ,
       -0.27942, -0.08309,  0.11655,  0.31154,  0.49411,
        0.65699,  0.79367,  0.89871,  0.96792,  0.99854,
        0.98936,  0.94073,  0.8546 ,  0.7344 ,  0.58492,
        0.41212,  0.22289,  0.02478, -0.17433, -0.36648,
       -0.54402], dtype=float32)
```

```
type(y_mix)
```

```
<class 'jaxlib._jax.ArrayImpl'>
```

JAX Arrays

As we've just seen a JAX array is very similar to a numpy array but there are some important differences.

- JAX arrays are immutable

```
x = jnp.array([3, 2, 1])
x[0] = 2
```

```
TypeError: JAX arrays are immutable and do not support in-place item assignment. Instead of x[idx]
```

- related to the above JAX does not support inplace operations - these functions now create and return a copy of the array

```
y = x.sort()
y
```

```
Array([1, 2, 3], dtype=int32)
```

```
x
```

```
Array([3, 2, 1], dtype=int32)
```

```
np.shares_memory(x,y)
```

```
False
```

- The default JAX array dtypes 32 bits not 64 bits (i.e. `float32` not `float64` and `int32` no

```
jnp.array([1, 2, 3])
```

```
Array([1, 2, 3], dtype=int32)
```

```
jnp.array([1, 2, 3], dtype=jnp.float64)
```

```
<string>:1: UserWarning: Explicitly requested dtype float64 requested in array is not available, and
Array([1., 2., 3.], dtype=float32)
```

64-bit dtypes can be enabled by setting `jax_enable_x64=True` in the JAX configuration

```
jax.config.update("jax_enable_x64", True)
```

```
jnp.array([1, 2, 3])
```

```
Array([1, 2, 3], dtype=int64)
```

- JAX arrays are allocated to one *or more* devices

```
jax.devices()
```

```
[CpuDevice(id=0)]
```

```
x.devices()
```

```
{CpuDevice(id=0)}
```

```
x.sharding
```

```
SingleDeviceSharding(device=CpuDevice(id=0
```

- Using JAX interactively allows for the use of standard Python control flow (`if`, `while`, `for`, etc.) but this is not supported for some of JAX's more advanced operations (e.g. `jit` and `grad`)

  There are replacements for most of these constructs in JAX, but they are beyond the scope of today.

Random number generation

JAX vs NumPy

Pseudo random number generation in JAX is a bit different than with NumPy - the latter depends on a global state that is updated each time a random function is called.

NumPy's PRNG guarantees something called sequential equivalence which amounts to sampling N numbers sequentially is the same as sampling N numbers at once (e.g. a vector of length N).

```
np.random.seed(0)
f"individually: {np.stack([np.random.uniform() for i in range(5)])}"
```

```
'individually: [0.54881 0.71519 0.60276 0.54488 0.42365]'
```

```
np.random.seed(0)
f"at once: {np.random.uniform(size=5)}"
```

```
'at once: [0.54881 0.71519 0.60276 0.54488 0.42365]'
```

Parallelization & sequential equivalence

Sequential equivalence can be problematic in when using parallelization across multiple devices, consider the following code:

```
np.random.seed(0)

def bar():
  return np.random.uniform()

def baz():
  return np.random.uniform()

def foo():
  return bar() + 2 * baz()
```

How do we guarantee that we get consistent results if we don't know the order that `bar()` and `baz()` will run?

PRNG keys

JAX makes use of *random keys* which are just a fancier version of random seeds - all of JAX's random functions require a key as their first argument.

```
key = jax.random.PRNGKey(1234); key
```

```
Array([   0, 1234], dtype=uint32)
```

```
jax.random.normal(key)
```

```
Array(-0.5402, dtype=float64)
```

```
jax.random.normal(key)
```

```
Array(-0.5402, dtype=float64)
```

```
jax.random.normal(key, shape=(3,))
```

```
Array([-0.5402 ,  0.43958, -0.01978], dtype=float64)
```

Note that JAX does not provide a sequential equivalence guarantee - this is so that it can support vectorization for the generation of PRN.

Splitting keys

Since a key is essentially a seed we do not want to reuse them (unless we want an identical output). Therefore to generate multiple different PRN we can split a key to deterministically generate two (or more) new keys.

```
key11, key12 = jax.random.split(key)
f"{key=}"
```

```
'key=Array([   0, 1234], dtype=uint32)'
```

```
f"{key11=}"
```

```
'key11=Array([1264997412, 2518116175], dtype=uin
```

```
f"{key12=}"
```

```
'key12=Array([2877103387, 1697627890], dtype=uin
```

```
key21, key22 = jax.random.split(key)
f"{key=}"
```

```
'key=Array([   0, 1234], dtype=uint32)'
```

```
f"{key21=}"
```

```
'key21=Array([1264997412, 2518116175], dtype=uin
```

```
f"{key22=}"
```

```
'key22=Array([2877103387, 1697627890], dtype=uin
```

```
key3 = jax.random.split(key, num=3)
key3
```

```
Array([[1264997412, 2518116175],
       [2877103387, 1697627890],
       [2113592192,  603280156]], dtype=uint32)
```

```
jax.random.normal(key, shape=(3,))
```

```
Array([-0.5402 ,  0.43958, -0.01978], dtype=float64)
```

```
jax.random.normal(key11, shape=(3,))
```

```
Array([ 1.24104,  0.12019, -2.2399 ], dtyp
```

```
jax.random.normal(key21, shape=(3,))
```

```
Array([ 1.24104,  0.12019, -2.2399 ], dtyp
```

```
jax.random.normal(key12, shape=(3,))
```

```
Array([ 0.07627, -1.3035 ,  0.86524], dtyp
```

```
jax.random.normal(key22, shape=(3,))
```

```
Array([ 0.07627, -1.3035 ,  0.86524], dtyp
```

```
jax.random.normal(key3[0], shape=(3,))
```

```
Array([ 1.24104,  0.12019, -2.2399 ], dtype=float64)
```

```
jax.random.normal(key3[1], shape=(3,))
```

```
Array([ 0.07627, -1.3035 ,  0.86524], dtype=float64)
```

```
jax.random.normal(key3[2], shape=(3,))
```

```
Array([-0.02894,  1.05075, -2.22082], dtype=float64)
```

JAX & jit

JAX performance

```
key = jax.random.PRNGKey(1234)
x_jnp = jax.random.normal(key, (1000,1000))
x_np = np.array(x_jnp)
```

```
%timeit y = x_np @ x_np
```

```
1.09 ms ± 92.4 µs per loop (mean ± std. dev. of 7 runs, 1,000 loops each)
```

```
%timeit y = x_jnp @ x_jnp
```

```
3.42 ms ± 122 µs per loop (mean ± std. dev. of 7 runs, 100 loops each)
```

```
%timeit y = 3*x_np + x_np
```

```
514 µs ± 41 µs per loop (mean ± std. dev. of 7 runs, 1,000 loops each)
```

```
%timeit y = 3*x_jnp + x_jnp
```

```
413 µs ± 24.2 µs per loop (mean ± std. dev. of 7 runs, 1,000 loops each)
```

jit

```
def SELU_np(x, α=1.67, λ=1.05):
  "Scaled Exponential Linear Unit"
  return λ * np.where(x > 0, x, α * np.exp(x
```

```
def SELU_jnp(x, α=1.67, λ=1.05):
  "Scaled Exponential Linear Unit"
  return λ * jnp.where(x > 0, x, α * jnp.exp
```

```
x = np.arange(1e6)
%timeit y = SELU_np(x)
```

```
4.08 ms ± 80 µs per loop (mean ± std. dev.
```

```
x = jnp.arange(1e6)
%timeit y = SELU_jnp(x)
```

```
1.58 ms ± 68.8 µs per loop (mean ± std. dev
```

```
SELU_np_jit = jax.jit(SELU_np)
```

```
SELU_jnp_jit = jax.jit(SELU_jnp)
```

```
%timeit y = SELU_np_jit(x)
```

```
TracerArrayConversionError: The numpy.ndarr
```

```
%timeit y = SELU_jnp_jit(x)
```

```
418 µs ± 13 µs per loop (mean ± std. dev. c
```

jit limitations

When it works the jit tool is fantastic, but it does have a number of limitations,

- Must use pure functions (no side effects)
- Must primarily use JAX functions
  - e.g. use `jnp.minimum()` not `np.minimum()` or `min()`
- Must generally avoid conditionals / control flow
- Issues around concrete values when tracing (static values)
- Check performance - there are not always gains + there is the initial cost of compilation

Automatic differentiation

Basics

The `grad()` function takes a numerical function, returning a scalar, and returns a function for calculating the gradient of that function.

```
def f(x):
  return x**2
```

```
f(3.)
```

```
9.0
```

```
jax.grad(f)(3.)
```

```
Array(6., dtype=float64, weak_t
```

```
jax.grad(
  jax.grad(f)
)(3.)
```

```
Array(2., dtype=float64, weak_t
```

```
def g(x):
  return jnp.exp(-x)
```

```
g(1.)
```

```
Array(0.36788, dtype=float64, w
```

```
jax.grad(g)(1.)
```

```
Array(-0.36788, dtype=float64,
```

```
jax.grad(
  jax.grad(g)
)(1.)
```

```
Array(0.36788, dtype=float64, w
```

```
def h(x):
  return jnp.maximum(0,x)
```

```
h(-2.)
```

```
Array(0., dtype=float64, weak_t
```

```
h(2.)
```

```
Array(2., dtype=float64, weak_t
```

```
jax.grad(h)(-2.)
```

```
Array(0., dtype=float64, weak_t
```

```
jax.grad(h)(2.)
```

```
Array(1., dtype=float64, weak_t
```

Aside - `vmap()`

I would like to plot `h()` and `jax.grad(h)()` - lets see what happens,

```
x = jnp.linspace(-3,3,101)
y = h(x)
y_grad = jax.grad(h)(x)
```

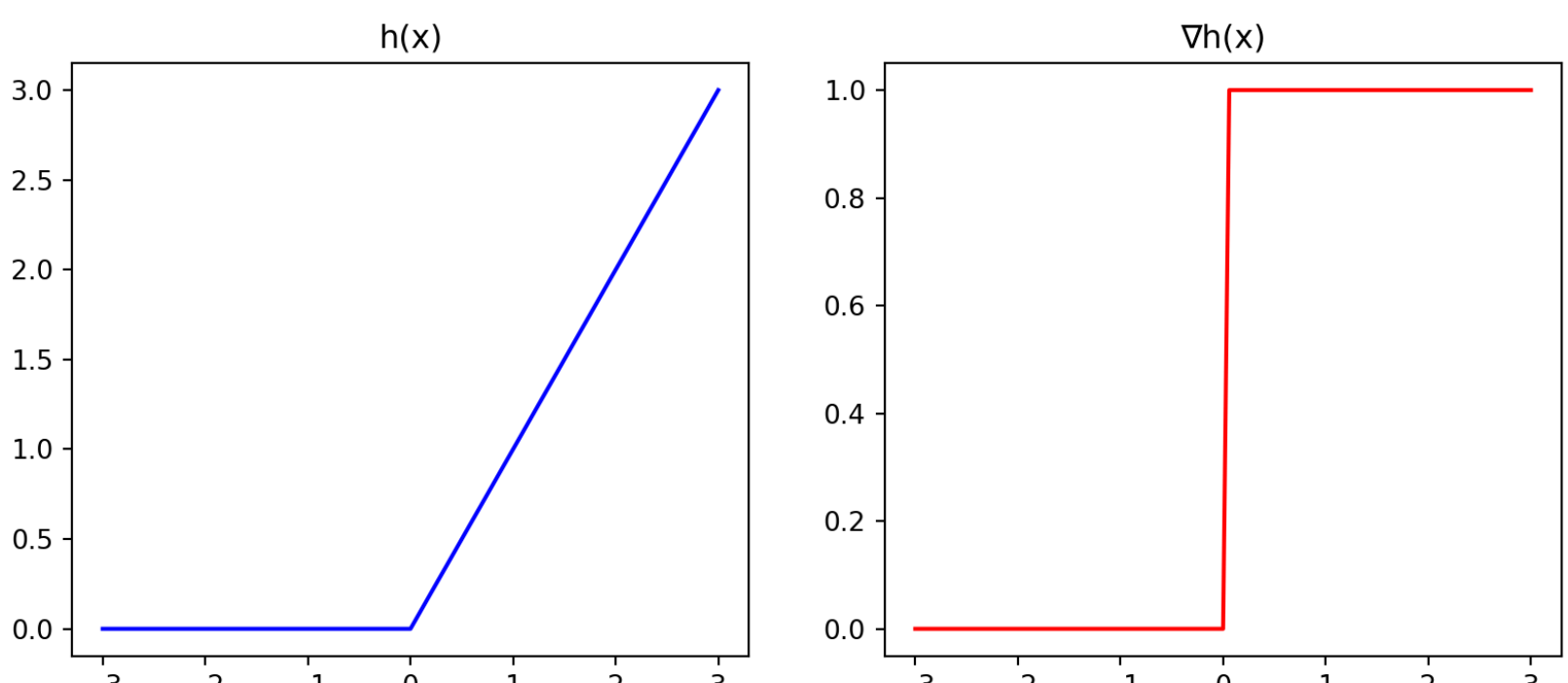
```
TypeError: Gradient only defined for scalar-output functions. Output had shape: (101,)
--------------------
For simplicity, JAX has removed its internal frames from the traceback of the followin
```

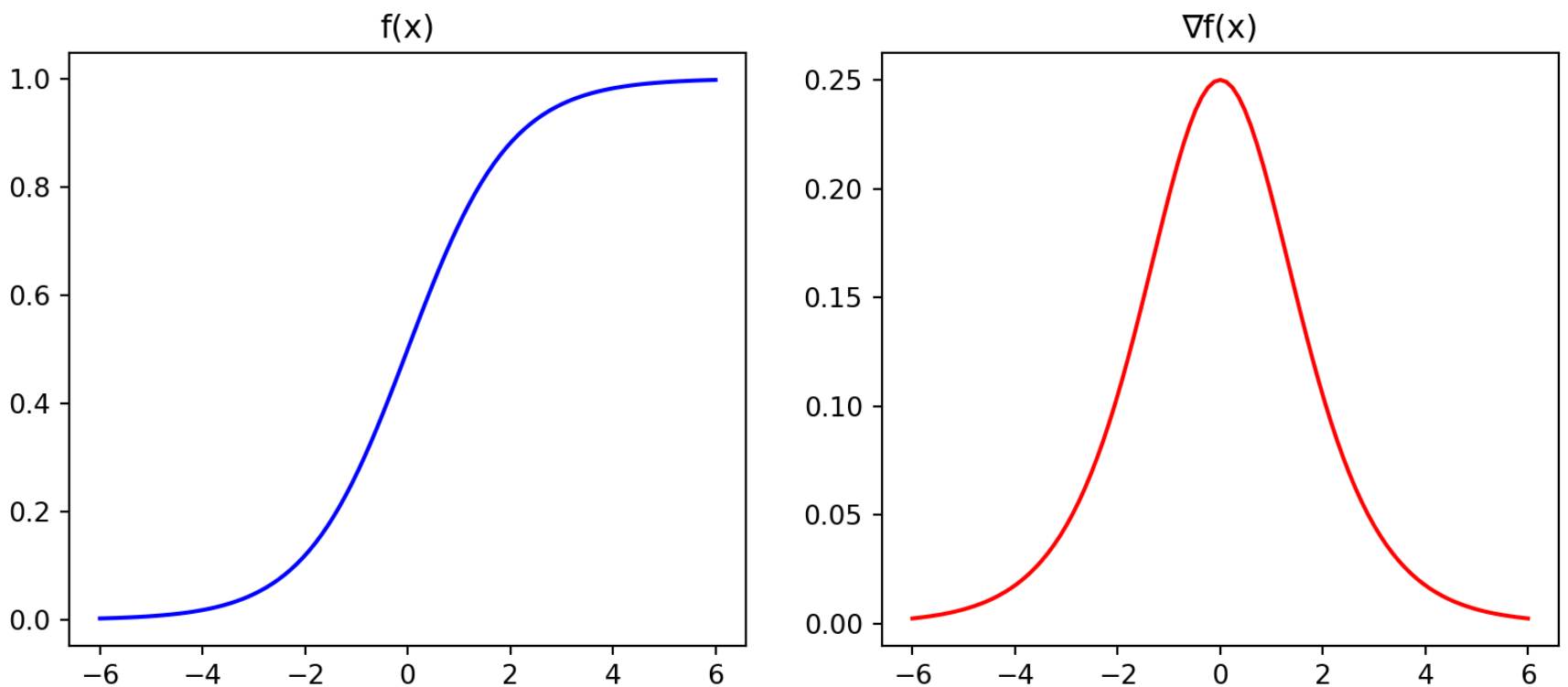
As just mentiond, we can only calculate the gradient for scalar valued functions. However, we can transform our scalar function into a vectorized function using `vmap()`.

```
h_grad = jax.vmap(
  jax.grad(h)
)
y_grad = h_grad(x)
```

```python
y_grad
```

```
Array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
       0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
       0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,
       0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 1.,
       1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
       1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
       1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1., 1.,
       1., 1., 1., 1., 1., 1., 1., 1., 1., 1.],      dtype=float64)
```



Another quick example

```
x = jnp.linspace(-6,6,101)
f = lambda x: 0.5 * (jnp.tanh(x / 2) + 1)
y = f(x)
y_grad = jax.vmap(jax.grad(f))(x)
```